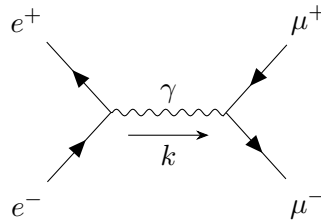


Quantum Field Theory

LECTURE NOTES

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Lecture 31: Feynman diagrams

In the previous case, we discussed about Wick's theorem and time-ordered product of fields can be simply written as a sum of contractions of the fields. We will now consider the case of ϕ^4 interacting theory and try to find the n -point correlation functions of the interacting theory.

Recall from Eq. ?? that the correlation functions are basically the vacuum expectation of the time-ordered field products and the time-ordered product is known from Wick's theorem. Thus we already have a recipe to calculate this and we will use a perturbative approach for this.

We had considered the interaction of the form ϕ^4 and from this $\mathcal{H}_I = \frac{\lambda}{4!}\phi^4$ where everything is in the interaction picture. Let us consider the **numerator** for the two-point correlation function first which is of the form

$$\langle \Omega | \mathcal{T} \left\{ \phi_1 \phi_2 \exp \left(\frac{-i\lambda}{4!} \int d^4 z \phi^4(z) \right) \right\} | \Omega \rangle$$

We first Taylor expand the exponential

$$\exp \left(\frac{-i\lambda}{4!} \int d^4 z \phi^4(z) \right) = 1 + \left(\frac{-i\lambda}{4!} \right) \int d^4 z \phi^4(z) + \left(\frac{-i\lambda}{4!} \right)^2 \int d^4 z d^4 w \phi^4(z) \phi^4(w) + \dots$$

and then the numerator takes on a ghastly form

$$\langle \Omega | \mathcal{T} \{ \phi_1 \phi_2 \} | \Omega \rangle + \left(\frac{-i\lambda}{4!} \right) \int d^4 z \langle \Omega | \mathcal{T} \{ \phi_1 \phi_2 \phi^4(z) \} | \Omega \rangle + \frac{1}{2!} \left(\frac{-i\lambda}{4!} \right)^2 \int d^4 z d^4 w \langle \Omega | \mathcal{T} \{ \phi_1 \phi_2 \phi^4(z) \phi^4(w) \} | \Omega \rangle + \dots$$

It is now possible to use Wick's theorem to characterise all these integrals, carefully considering the cumbersome contractions.

▷ Term of $\mathcal{O}(1)$:

The term is $\langle \Omega | \mathcal{T} \{ \phi_1 \phi_2 \} | \Omega \rangle$, which is simply Δ_{12} and is denoted diagrammatically as



▷ Term of $\mathcal{O}(\lambda)$:

We have to work a little bit here. The term is

$$\int d^4 z \langle \Omega | \mathcal{T} \{ \phi_1 \phi_2 \phi^4(z) \} | \Omega \rangle \equiv \int d^4 z \langle \Omega | \mathcal{T} \{ \phi_1 \phi_2 \phi(z) \phi(z) \phi(z) \phi(z) \} | \Omega \rangle$$

We have to find all type of contractions amidst these six fields if we use the Wick's theorem. Note that any partial contractions will not matter, since we are taking vacuum expectations and uncontracted normal ordered products will kill it. Thus we need to consider only fully contracted fields. How many such configurations are there?

It turns out that we need to do some combinatorics for this. Indeed, all *good* math and physics ends in either linear algebra or combinatorics¹.

So if we take one field, then it can be contracted with the remaining 5, then these two fields are gone, remaining four are there. If we choose one, we can contract with remaining 3, and then the last two automatically gets contracted. So total $5 \times 3 = 15$ such fully contracted field configurations.

Due to symmetry conditions, we only have two distinct types of contractions which are,

For the first case, we have $x - y$ contracting with each other and for the second case, we have $x - z$ contracting. These can be diagrammatically represented as:

¹which implies that if you don't find yourself doing these, you are not doing good physics

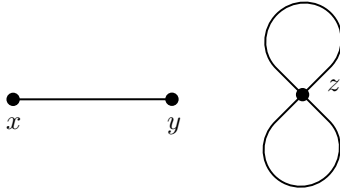


Figure 1: Type I contraction

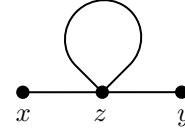


Figure 2: Type II contraction

There are three of the above Type I terms, since if x contracts with y then the first z can contract with three other guys.

Let us consider the second type of contraction where x contracts with z . There are 12 such terms since x can go with z in 4 ways and for each xz pair there are three ways y can go with z . Then the term of order $\mathcal{O}(\lambda)$ becomes

$$3 \times \left(\frac{-i\lambda}{4!} \right) \int d^4z \Delta^2(x-y) \Delta^2(z-z) + 12 \times \left(\frac{-i\lambda}{4!} \right) \int d^4z \Delta(x-z) \Delta(y-z) \Delta(z-z)$$

(For now just take $\Delta(z-z)$ to be some ‘number’ (even though it diverges) and let us not worry about this!)

Some observations

- ▷ The outermost nodes in the diagram (Eg. x and y) are the *external vertices*. These vertices are not integrated over.
- ▷ The inner nodes (Eg. z) are called the *internal vertices* and for each internal vertex, we have to associate an integral to them.

We had found the numerator upto order λ and now let us find the **denominator**.

$$1 + \left(\frac{-i\lambda}{4!} \right) \int d^4z \langle \Omega | \mathcal{T} \{ \phi^4(z) \} | \Omega \rangle + \frac{1}{2!} \left(\frac{-i\lambda}{4!} \right)^2 \int d^4z d^4w \langle \Omega | \mathcal{T} \{ \phi^4(z) \phi^4(w) \} | \Omega \rangle + \dots$$

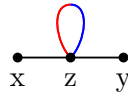
Note that the denominator does not have any free fields ϕ_1 and ϕ_2 and thus, the $\mathcal{O}(1)$ the term will be just 1 while the term of $\mathcal{O}(\lambda)$ will consist of only the contractions with z (3 terms). Thus symbolically (since the integrals will be just some number) we can write the two point correlator function as:

$$\begin{aligned}
 & \frac{
 \begin{array}{c}
 \text{---} \bullet \text{---} \bullet \text{---} \\
 x \qquad y
 \end{array}
 + 3 \left(\begin{array}{c} \text{---} \bullet \text{---} \text{---} \text{---} \text{---} \bullet \text{---} \end{array} \right)
 + 12 \left(\begin{array}{c} \text{---} \bullet \text{---} \text{---} \text{---} \text{---} \bullet \text{---} \end{array} \right)
 + \mathcal{O}(\lambda^2)
 }{
 1 + 3 \text{---} \bullet \text{---} \text{---} \text{---} \text{---} \bullet \text{---} + \mathcal{O}(\lambda^2)
 } \\
 &= \left[\begin{array}{c} \text{---} \bullet \text{---} \bullet \text{---} \\ x \qquad y \end{array} + 3 \left(\begin{array}{c} \text{---} \bullet \text{---} \text{---} \text{---} \text{---} \bullet \text{---} \end{array} \right) + 12 \left(\begin{array}{c} \text{---} \bullet \text{---} \text{---} \text{---} \text{---} \bullet \text{---} \end{array} \right) + \mathcal{O}(\lambda^2) \right] \left[1 - 3 \text{---} \bullet \text{---} \text{---} \text{---} \text{---} \bullet \text{---} + \mathcal{O}(\lambda^2) \right] \\
 &= \begin{array}{c} \text{---} \bullet \text{---} \bullet \text{---} \\ x \qquad y \end{array} - 3 \left(\begin{array}{c} \text{---} \bullet \text{---} \text{---} \text{---} \text{---} \bullet \text{---} \end{array} \right) + 3 \left(\begin{array}{c} \text{---} \bullet \text{---} \text{---} \text{---} \text{---} \bullet \text{---} \end{array} \right) + 12 \left(\begin{array}{c} \text{---} \bullet \text{---} \text{---} \text{---} \text{---} \bullet \text{---} \end{array} \right) + \mathcal{O}(\lambda^2) \\
 &= \left[\begin{array}{c} \text{---} \bullet \text{---} \bullet \text{---} \\ x \qquad y \end{array} + 12 \left(\begin{array}{c} \text{---} \bullet \text{---} \text{---} \text{---} \text{---} \bullet \text{---} \end{array} \right) + \mathcal{O}(\lambda^2) \right]
 \end{aligned}$$

Note than, a cancellation was involved in the reduction such that the diagram with two disconnected components got cancelled. Even though this seems like a miracle, in fact, it's not!

Turns out, only those diagrams contribute where the components are connected to the external points. The contributions from all the disconnected subgraphs cancel. Before showing this, let us write a summary of what we need to do for systematic calculations:

- For each line  we associate the Feynman propagator $\Delta(x - y)$.
- For each internal vertex  we associate the term $(-i\lambda) \int d^4z$ (that is, we integrate over any internal vertices).
- For each external vertex, we associate the value 1.
- Finally we divide by the *symmetry factor*, which is like an *enigmatic term* but basically means the number of ways we can rearrange the propagators or vertices, without changing the diagram (Peskin says that we *almost surely* never need to calculate any diagram with symmetry factor greater than 2). For example, in the diagram



we can interchange the blue and the red lines (propagators) which would keep the diagram intact. Hence there are two ways of arrangement and hence, the symmetry factor with which we divide the expression is 2.

This is consistent with our explicit calculation, since we get a contribution of 12 for this diagram and then we divide by the $4!$ which is already there in the integral, giving us $12/24 = 1/2$.

The above set of rules are called the *Feynman rules* in position-space (since we are considering the fields in the real space) and these are specific to the ϕ^4 -theory. For other type of interactions, few changes are accordingly made.

Theorem 1 (Linked-Cluster Theorem):

$$\langle \tilde{\Omega} | \mathcal{T} \{ \phi_1 \dots \phi_n \} | \tilde{\Omega} \rangle = \left(\begin{array}{l} \text{sum of all connected diagrams} \\ \text{subjected to Feynman rules} \end{array} \right)$$

Proof. A typical diagram consists of *connected* and *disconnected* diagrams, where connected diagrams are the ones where everything is connected to some external vertices. Disconnected diagrams are such that the diagrams consist of different components where at least one component is not connected to any external vertex. The components not connected to any external vertices are called the *vacuum bubbles*.

Let us consider the set of all vacuum bubbles represented by $\mathcal{V}$, then we have

$$\mathcal{V} = \left\{ \text{figure-eight} , \text{two figure-eights} , \text{figure-eight with a bubble} , \text{etc.} \right\}$$

Suppose a diagram V (where V actually denotes the value of the integral represented by the diagram) has n_i pieces of $v_i \in \mathcal{V}$, in addition to the connected components (containing external vertices) whose value is, let say, c . Then the contribution of the diagram becomes:

$$c \cdot \prod_i \frac{1}{n_i!} (v_i)^{n_i}$$

where the $n_i!$ is taken since there are n_i copies of v_i , but these copies are indistinguishable. Then the numerator of the correlation function is obtained by summing over all the different connected components

c

$$\text{numerator} = \sum_{\substack{\text{all possible} \\ \text{connected} \\ \text{pieces } c}} \sum_{\{n_i\}} c \cdot \prod_i \frac{1}{n_i!} (v_i)^{n_i} = \sum_{\substack{\text{all possible} \\ \text{connected} \\ \text{pieces } c}} c \cdot \prod_i \sum_{\{n_i\}} \frac{1}{n_i!} (v_i)^{n_i} = \left(\sum_{\substack{\text{all possible} \\ \text{connected} \\ \text{pieces } c}} c \right) \cdot \left(\exp \left(\sum_i v_i \right) \right)$$

where we have used the expansion for e^x and we finally obtain that the numerator is just the product of the connected contribution and the vacuum bubble contribution. The denominator has no connected contributions since the external fields do not occur in the denominator. Hence,

$$\text{denominator} = \exp \left(\sum_i v_i \right)$$

The exponential part thus cancels when we divide the numerator and the denominator, leaving only the contribution from the connected components.

NOTE: Higher order (> 2 point) correlation functions may have apparent disconnected pieces in the diagram, like say, the diagrams for the four point correlation function in ??, where two of the diagrams are not connected. However, each disconnected piece is itself connected to some or the other external vertices. When we talk about disconnected pieces (or vacuum bubbles), we talk about diagrams which contain a piece where that piece has no connection with any of the external vertex.

Lecture 32: Feynman diagrams in Momentum space

Recall that the Feynman propagator and its Fourier transform were defined as

$$D_F(x-y) = \int \frac{d^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} e^{-ik \cdot (x-y)} \quad \tilde{D}_F(k) = \frac{i}{k^2 - m^2 + i\epsilon}$$

Then, if we convert the correlation function to the momentum space, we would have product of these propagators and also, we would introduce a delta function for any internal vertex. For example,

$$\begin{aligned} \int d^4z \Delta(x-z) \Delta(y-z) &\sim \int d^4z \frac{d^4p_1}{(2\pi)^4} \frac{d^4p_2}{(2\pi)^4} \tilde{D}_F(p_1) \tilde{D}_F(p_2) e^{-ip_1 \cdot x - ip_2 \cdot y} e^{-i(p_1+p_2) \cdot z} \\ &\sim \int \frac{d^4p_1}{(2\pi)^4} \frac{d^4p_2}{(2\pi)^4} \tilde{D}_F(p_1) \tilde{D}_F(p_2) \delta^{(4)}(p_1 + p_2) e^{-ip_1 \cdot x - ip_2 \cdot y} \end{aligned}$$

Note that the delta function in some sense, denotes *momentum conservation*. The Feynman rules for the correlation function in momentum space are as follows:

- For each internal line, we associate a momentum k and the propagator $\tilde{D}_F(k)$
- For each vertex, we write down the factor $(-i\lambda)(2\pi)^4 \delta^{(4)}\left(\sum_i k_i\right)$ where $\sum_i k_i$ is the sum of all momenta flowing *into* the vertex that we are considering (*momentum conservation*).
- Divide by the symmetry factor and the other things that we did before.

The best thing would be to demonstrate through an example. Consider the diagram,

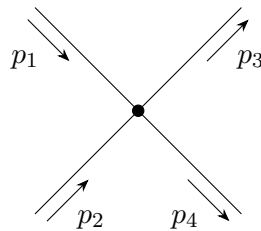


Figure 3: Diagram of order $\mathcal{O}(\lambda)$ for the four-point correlation function with interaction.

We can put the arrows of the momentum in any direction for now, the second rule will incorporate the sign automatically. The contribution of this diagram to the four-point correlation function, according to the above rules would be,

$$(-i\lambda)(2\pi)^4 \left(\prod_{j=1}^4 \frac{i}{p_j^2 - m^2 + i\epsilon} \right) \delta^{(4)}(p_1 + p_2 - p_3 - p_4)$$

Now, consider a bit more involved diagram, containing a loop.

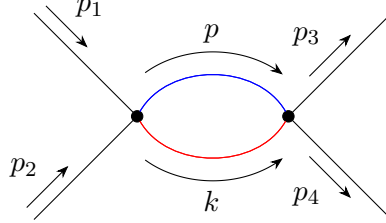


Figure 4: Diagram of order $\mathcal{O}(\lambda^2)$, containing a loop, for the four-point correlation function with interaction.

Let us first impose the momentum conservation, which will give us $\delta^{(4)}(p_1 + p_2 - p - k)$ and $\delta^{(4)}(p + k - p_3 - p_4)$. Note that, these delta functions will be in product with each other. The argument of the second delta function becomes zero for $p = p_3 + p_4 - k$ and we can substitute this in the first delta function, leading to $\delta^{(4)}(p_1 + p_2 - p_3 - p_4)$.

We see that, both p and k cannot be uniquely determined just from momentum conservation and hence, there is always a dangling momentum remaining, which we need to integrate out.

Also, we can interchange the red and blue lines, leaving the diagram unchanged, giving us a symmetry factor of 2. Thus, considering all the facts, the contribution of the diagram becomes:

$$\frac{1}{2}(-i\lambda)^2(2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - p_4) \int \frac{d^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} \frac{i}{(p_3 + p_4 - k)^2 - m^2 + i\epsilon} \left(\prod_{j=1}^4 \frac{i}{p_j^2 - m^2 + i\epsilon} \right)$$

2.1 Scattering Matrix

We now focus on how QFT can be used to extract information from physical processes, particularly *scattering*. Consider a state $|i\rangle$ which is at $t \rightarrow -\infty$, describing particles, which evolves in time and goes to a final state $|f\rangle$ at $t \rightarrow +\infty$.

In the initial and final states, all the particles are out of each others' influence, however, in an intermediate time, these come together and *interact*. The amplitude for this process is given by:

$$\lim_{\tau \rightarrow \infty} \langle f | U(\tau, -\tau) | i \rangle \equiv \langle f | \mathcal{S} | i \rangle$$

where $\mathcal{S}$ is called the *scattering matrix* or the *S-matrix*. $\mathcal{S}$ is an unitary operator. In almost all cases, we will neglect the situation where no scattering has occurred, in which case we can define

$$\mathcal{S} = 1 + i\mathcal{T}$$

and we will focus on the transfer matrix $\mathcal{T}$ henceforth. Till now we have been using the correlation function for all our interactions, however, it seems that the scattering amplitude is a more suitable quantity to measure in experimental observations.

There exists a thing called **Lehmann-Symanzik-Zimmermann (LSZ)** reduction formula which relates the correlation function with the scattering amplitude. The proof seems to be very complicated (for me), hence I am just going to mention the idea behind it.

Basically, for scattering amplitude, you are really interested in the actual *interaction* however, the correlation function also contains the free propagators.

What I mean is that, consider the diagram with the loop. For scattering processes, we generally know the initial and final momenta (momenta with which the particles are entering or going out), so we do not need to care about them. Thus, from the contribution, we remove the propagators of these ‘known’ momenta. This process is called *amputation*, where we remove the contribution from the external legs and this gives us the scattering matrix element.

There is always a delta function imposing momentum conservation which might become cumbersome to write everytime, hence we define the matrix $\mathcal{M}$ such that,

$$\langle \mathbf{p}_1 \dots \mathbf{p}_n | iT | \mathbf{k}_1 \dots \mathbf{k}_m \rangle = (2\pi)^4 \delta^{(4)} \left(\sum_i p_i - \sum_j k_j \right) (i\mathcal{M})$$

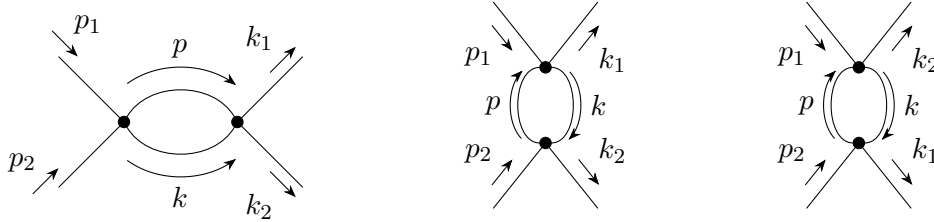
where p_i and k_j are the final and initial momenta. The matrix $\mathcal{M}$ will have contributions from the non-external propagators of all the diagrams that is possible for the process. In a gist, we have

$$\boxed{\mathcal{M} \equiv \text{sum of all connected, amputated diagrams}}$$

For example, consider a process

$$p_1 + p_2 \longrightarrow k_1 + k_2$$

Apart from the $\mathcal{O}(\lambda)$ diagram shown before, there are three order $\mathcal{O}(\lambda^2)$ diagrams which can represent the above scattering process and we have to consider all three of them:



- In the first diagram, momentum conservation gives $p_1 + p_2 = p + k = p_3 + p_4 \implies p = p_1 + p_2 - k$
- In the second diagram, momentum conservation gives $p_1 + p = k_1 + k \implies p = k_1 + k - p_1$
- In the first diagram, momentum conservation gives $p_1 + p = k_2 + k \implies p = k_2 + k - p_1$

The $\mathcal{O}(\lambda)$ process will contribute a factor of only $-i\lambda$ in the scattering matrix, since it has no loops (such a diagram is called a *tree-level diagram*) and hence no undetermined momenta to integrate out. Let us define the following quantity,

$$\mathcal{A}(p) \equiv \frac{(-i\lambda)^2}{2} \int \frac{d^4 k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} \frac{i}{(p - k)^2 - m^2 + i\epsilon}$$

Using $\mathcal{A}(p)$, we can write the matrix $\mathcal{M}$ upto $\mathcal{O}(\lambda^2)$ for the scattering process $p_1 + p_2 \longrightarrow k_1 + k_2$:

$$i\mathcal{M} = \underbrace{(-i\lambda)}_{\text{tree-level}} + \mathcal{A}(p_1 + p_2) + \mathcal{A}(p_1 - k_1) + \mathcal{A}(p_1 - k_2)$$

Lecture 33: Scalar Yukawa Theory

We have dealt with ϕ^4 interaction in the previous few sections and we saw some Feynman diagrams for the interaction. Now we will consider the interaction between a real scalar and a complex scalar field, which goes by the name of *Yukawa interaction* which is of the form:

$$\mathcal{H} = g\psi^\dagger\psi\phi$$

where ψ is our charged complex field and ϕ is the real scalar field. The full Lagrangian is given by:

$$\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{1}{2}m^2\phi^2 + (\partial_\mu\psi^\dagger)(\partial^\mu\psi) - M^2\psi^\dagger\psi - g\psi^\dagger\psi\phi$$

where m is the mass associated with the ϕ field and M is the mass associated with the ψ field. The $1/2$ factor does not come with the complex scalar field, since a complex scalar field already has two degrees of freedom. And regarding the interaction, the simplest case is considered. $\psi^\dagger\psi$ gives a real, Lorentz invariant quantity and we multiply it with ϕ to get us an interaction between the two fields.

Having been familiarised with Feynman diagrams before, we will straightaway jump to calculate the amplitude of some scattering processes for this interaction. Before that we need to consider a few things:

- We have two kinds of field here, so we need a way to distinguish between them. We will denote the real scalar field by *dashed lines* while the complex scalar field by *solid lines*. Also, the solid lines will be *directed*, to denote the charge flow of the complex scalar field.

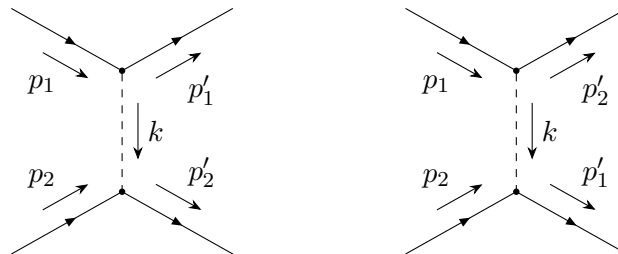
The convention will be to have an *incoming* arrow for the initial state in ψ . For the initial state in $\psi^\dagger$, we will have an *outgoing* arrow. For the final state, the convention is reversed.

- Each interaction vertex will have *three* fields, since the interaction term has three fields. In ϕ^4 theory, we have tetravalent interaction vertices since the interaction had four terms in ϕ .
- The Feynman rules mostly remain same in this case. The propagator for real and complex field is the same, only the appropriate mass of the field needs to be taken in each case. Each vertex will have a factor of $(-ig)$.

3.1 Nucleon-Nucleon scattering

Consider the process $\psi\psi \longrightarrow \psi\psi$ where two incoming ψ particles result in two outgoing ψ particles. Henceforth, we will call ψ particles are *nucleons* and ϕ particles as *mesons*. Hence this is a case of nucleon-nucleon scattering.

At order $\mathcal{O}(g^0)$ we have no scattering, both ψ comes and leaves as it is, without any interaction. Diagram of order $\mathcal{O}(g)$ with only one vertex is not possible for this process. Hence, the only meaningful diagram will start at order $\mathcal{O}(g^2)$. The simplest diagrams contributing to this process are,



Explicitly written with the momenta, this scattering process reads

$$\psi(p_1) + \psi(p_2) \longrightarrow \psi(p'_1) + \psi(p'_2)$$

Consider the first diagram. Momentum conservation leads to $k = p_1 - p'_1 = p'_2 - p_2$. Since there are two vertices, the contribution of the vertex will be $(-ig)^2$. We will consider only the propagator for k , since all the other momenta are determined. Then the scattering amplitude for the first diagram will be,

$$i\mathcal{M}_1 = (-ig)^2 \frac{i}{(p_1 - p'_1)^2 - m^2 + i\epsilon}$$

where we have substituted $k = p_1 - p'_1$ and since k is associated with the ϕ field, we considered the mass m . Similarly, for the second diagram we have,

$$i\mathcal{M}_2 = (-ig)^2 \frac{i}{(p_1 - p'_2)^2 - m^2 + i\epsilon}$$

The total scattering amplitude is the sum of these two terms,

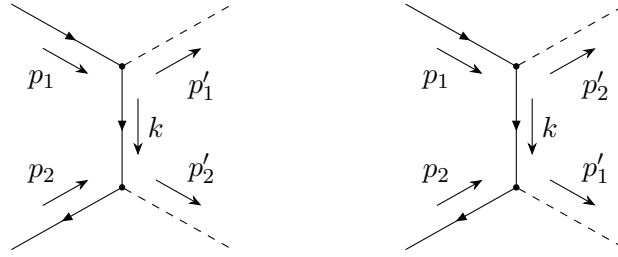
$$i\mathcal{M} = (-ig)^2 \left[\frac{i}{(p_1 - p'_1)^2 - m^2 + i\epsilon} + \frac{i}{(p_1 - p'_2)^2 - m^2 + i\epsilon} \right]$$

These diagrams are tree-level diagrams and have no loops and hence no undetermined momentum to integrate over. The loops come in at $\mathcal{O}(g^4)$ and higher orders.

3.2 Nucleon to Meson Scattering

Let us consider another process,

$$\psi(p_1) + \psi^\dagger(p_2) \longrightarrow \phi(p'_1) + \phi(p'_2)$$



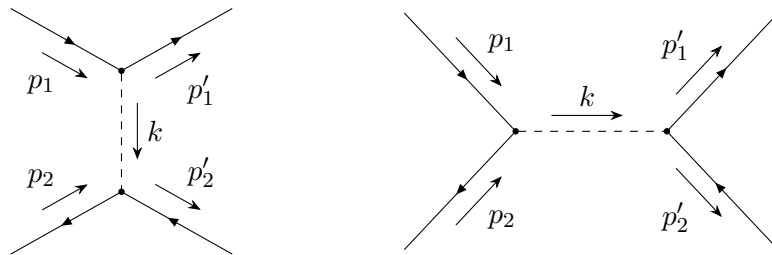
Observe the direction of the arrows denoting the charges. These have been drawn taking care of the *charge conservation* at each vertices. Since the internal line is now a *nucleon*, the amplitude is,

$$i\mathcal{M} = (-ig)^2 \left[\frac{i}{(p_1 - p'_1)^2 - M^2 + i\epsilon} + \frac{i}{(p_1 - p'_2)^2 - M^2 + i\epsilon} \right]$$

3.3 Nucleon-Anti-Nucleon Scattering

Consider yet another process, which will have a bit different diagram than the previous two

$$\psi(p_1) + \psi^\dagger(p_2) \longrightarrow \psi(p'_1) + \psi^\dagger(p'_2)$$



The second diagram is a tad different from the other diagrams that we have seen. Momentum conservation in the second diagram gives $p_1 + p_2 = k$. Thus the amplitude for the process will be given by,

$$i\mathcal{M} = (-ig)^2 \left[\frac{i}{(p_1 - p'_1)^2 - m^2 + i\epsilon} + \frac{i}{(p_1 + p_2)^2 - m^2 + i\epsilon} \right]$$

3.4 Mandelstam variables

The quantities $(p_1 - p'_1)$, $(p_1 - p'_2)$ and $(p_1 + p_2)$ seems to appear in the propagator expression quite often. These are given some standard names, called *Mandelstam variables*, which are defined as the Lorentz invariant quantities,

$$\begin{aligned}s &= (p_1 + p_2)^2 = (p'_1 + p'_2)^2 \\ t &= (p_1 - p'_1)^2 = (p_2 - p'_2)^2 \\ u &= (p_1 - p'_2)^2 = (p_2 - p'_1)^2\end{aligned}$$

These are also called s, t, u -channels. Firstly note that we have the total momentum conservation, $p_1 + p_2 = p'_1 + p'_2$. Then,

$$\begin{aligned}s + t + u &= (p_1 + p_2)^2 + (p_1 - p'_1)^2 + (p_1 - p'_2)^2 \\ &= p_1^2 + p_2^2 + 2p_1 \cdot p_2 + p_1^2 + p_1'^2 - 2p_1 \cdot p'_1 + p_1^2 + p_2'^2 - 2p_1 \cdot p'_2 \\ &= 3p_1^2 + p_2'^2 + p_1'^2 + p_2^2 + 2p_1 \cdot (p_2 - p'_1 - p'_2) \\ &= 3p_1^2 + p_2'^2 + p_1'^2 + p_2^2 - 2p_1^2 \\ &= p_1^2 + p_2'^2 + p_1'^2 + p_2^2\end{aligned}$$

where $p_i^2 \equiv (p_i)^\mu (p_i)_\mu$ and $p_i \cdot p_j \equiv (p_i)^\mu (p_j)_\mu$ are the Lorentz scalar product. Using the on-shell condition, $p_j^2 = m_j^2$ we have $s + t + u = \sum m_j^2$, which is a nice thing perhaps. Now, consider the *centre of mass/momentum* frame, where the total three-momentum is zero. Then,

$$\begin{aligned}\mathbf{p}_1 + \mathbf{p}_2 &= 0 \implies \mathbf{p}_i \equiv \mathbf{p}_1 = -\mathbf{p}_2 \\ \mathbf{p}'_1 + \mathbf{p}'_2 &= 0 \implies \mathbf{p}_f \equiv \mathbf{p}'_1 = -\mathbf{p}'_2\end{aligned}$$

From the above conditions, the momentum four vectors become

$$p_1 = (E_1, \mathbf{p}_i), p_2 = (E_2, -\mathbf{p}_i), p'_1 = (E'_1, \mathbf{p}_f), p'_2 = (E'_2, -\mathbf{p}_f)$$

Then, $s = (p_1 + p_2)^2 \equiv (E_1 + E_2)^2 \equiv E_{\text{T}}^2$. s thus measures the total center of mass energy of the collision.

We can now find an expression for the energies in terms of the variable s . For that, we will require the on-shell condition, $p^2 = m^2$ and the above relation, which gives us:

$$E_1 + E_2 = \sqrt{s} \quad E_1^2 - \mathbf{p}_i^2 = m_1^2 \quad E_2^2 - \mathbf{p}_i^2 = m_2^2$$

Using the above, we get the following,

$$m_1^2 - m_2^2 = E_1^2 - E_2^2 = E_1^2 - (\sqrt{s} - E_1)^2 = \sqrt{s}(2E_1 - \sqrt{s}) \implies E_1 = \frac{s + m_1^2 - m_2^2}{2\sqrt{s}}$$

Similarly, we obtain the other energies in terms of the masses and s ,

$$E_2 = \frac{s + m_2^2 - m_1^2}{2\sqrt{s}} \quad E'_1 = \frac{s + m_1'^2 - m_2'^2}{2\sqrt{s}} \quad E'_2 = \frac{s + m_2'^2 - m_1'^2}{2\sqrt{s}}$$

Let us now consider the variable $t = (p_1 - p'_1)^2$ which turns out to be

$$\begin{aligned}t &= p_1^2 + p_1'^2 - 2p_1 \cdot p'_1 \\ &= p_1^2 + p_1'^2 - 2((p_1)^0 (p'_1)^0 - |\mathbf{p}_1| |\mathbf{p}'_1| \cos \theta) \\ &= p_1^2 + p_1'^2 - 2(E_1 E'_1 - |\mathbf{p}_1| |\mathbf{p}'_1| \cos \theta)\end{aligned}$$

where θ is the angle between $\mathbf{p}_1$ and $\mathbf{p}'_1$. Let us take the elastic limit, $m_1 = m'_1$ and $m_2 = m'_2$ which gives us $E_1 = E'_1 = \mathfrak{E}_1$ and $E_2 = E'_2 = \mathfrak{E}_2$ along with $|\mathbf{p}_i| = |\mathbf{p}_f| \equiv \mathbf{p}$. Then the expression for t becomes,

$$t = \cancel{2\mathfrak{E}_1^2} - 2\mathbf{p}^2 - \cancel{2\mathfrak{E}_1^2} + 2\mathbf{p}^2 \cos \theta = -2\mathbf{p}^2(1 - \cos \theta) = -4\mathbf{p}^2 \sin^2 \left(\frac{\theta}{2} \right) \leq 0$$

In a similar way, we can derive $u = -4\mathbf{p}^2 \cos^2\left(\frac{\theta}{2}\right) \leq 0$. Thus, t and u are measures of the momentum exchanged between particles. As the Mandelstam variables are scalars, these will retain these properties in all frames, so $s \geq 0$ and $t, u \leq 0$ in all frames.

The amplitudes of the scattering processes can be written entirely in terms of the Mandelstam variables.

- For the process $\psi\psi \rightarrow \psi\psi$ we have $i\mathcal{M} = (-ig)^2 \left[\frac{i}{t-m^2} + \frac{i}{u-m^2} \right]$
- For the process $\psi\psi^\dagger \rightarrow \phi\phi$ we have $i\mathcal{M} = (-ig)^2 \left[\frac{i}{t-M^2} + \frac{i}{u-M^2} \right]$
- For the process $\psi\psi^\dagger \rightarrow \psi\psi^\dagger$ we have $i\mathcal{M} = (-ig)^2 \left[\frac{i}{t-m^2} + \frac{i}{s-m^2+i\epsilon} \right]$

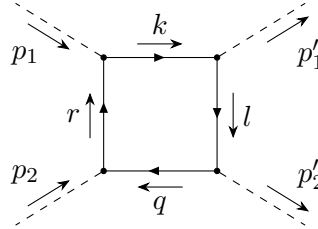
As t and u are always negative, the denominator in the amplitudes (except the red term) cannot be zero and hence we have safely removed the $+i\epsilon$ term. However, the red term with s channel can be zero, as $s \geq 0$. Take the COM frame for the s channel diagram. Since the particles are all nucleons, $s = E_T^2 = 4(M^2 + \mathbf{p}^2)$ and hence the denominator becomes $4(M^2 + \mathbf{p}^2) - m^2$. If $m < 2M$ then $4M^2 - m^2 > 0$ which implies that the denominator is positive, so we can drop the $+i\epsilon$ term. If $m > 2M$ then the amplitude corresponding to the second diagram diverges at some value of the momentum and we have to keep $+i\epsilon$.

3.5 Meson Scattering

This will be another different process with a different diagram.

$$\phi(p_1) + \phi(p_2) \rightarrow \phi(p'_1) + \phi(p'_2)$$

There are two incoming and two outgoing ϕ lines and we have to put two ψ lines to each of them. If we think about this, the simplest diagram for this process will contain a loop (which increases the complexity)!



Since there is a loop, all the momentum cannot be uniquely determined solely from momentum conservation. Let us keep l fixed and try to express k, r, q in terms of l . Momentum conservation at each vertex gives:

$$\triangleright \boxed{k = l + p'_1} \quad \triangleright l = q + p'_2 \implies \boxed{q = l - p'_2} \quad \triangleright p_1 + r = k = l + p'_1 \implies \boxed{r = l + p'_1 - p_1}$$

The scattering amplitude will have the integral over the undetermined momentum l and the product of propagators containing l . Since there are four vertices, there will be a contribution of $(-ig)^4$

$$(-ig)^4 \int \frac{d^4k}{(2\pi)^4} \frac{i}{[l^2 - M^2 + i\epsilon]} \frac{i}{[(l + p'_1)^2 - M^2 + i\epsilon]} \frac{i}{[(l - p'_2)^2 - M^2 + i\epsilon]} \frac{i}{[(l + p'_1 - p_1)^2 - M^2 + i\epsilon]}$$

The integral seems to be complicated. I don't know how to do this. Even Tong has left the integral like this, so it's a good time for me to stop too!

There are two other diagrams contributing to the above process which we will see later. Another thing to say is that, these amplitudes can also be *painstakingly* found out using Dyson formula and Wick's theorem, however, Feynman rules makes it really easy and elegant. And that is why we all love him!¹

¹The other set of more exciting Feynman rules about life can be found in his book *Surely You're Joking, Mr. Feynman!*

Lecture 33: Spinor Yukawa Theory

We have seen interactions only between scalar fields upto now. We had also seen fermion fields and gauge fields earlier. Hence we will discuss what happens when fermions get coupled to scalar fields, which is the actual *Yukawa theory*. The interaction is of the form

$$\mathcal{H}_{\text{int}} = g\phi\bar{\Psi}\Psi$$

where Ψ is a Dirac spinor and $\bar{\Psi}$ is the Dirac adjoint. We needed to make a Lorentz invariant scalar and hence the simplest interaction, as seen before, had to be of this form. The entire Lagrangian becomes

$$\mathcal{L} = \frac{1}{2}(\partial^\mu\phi)(\partial_\mu\phi) - \frac{1}{2}m^2\phi^2 + \bar{\Psi}(i\not{\partial} - M)\Psi - g\phi\bar{\Psi}\Psi$$